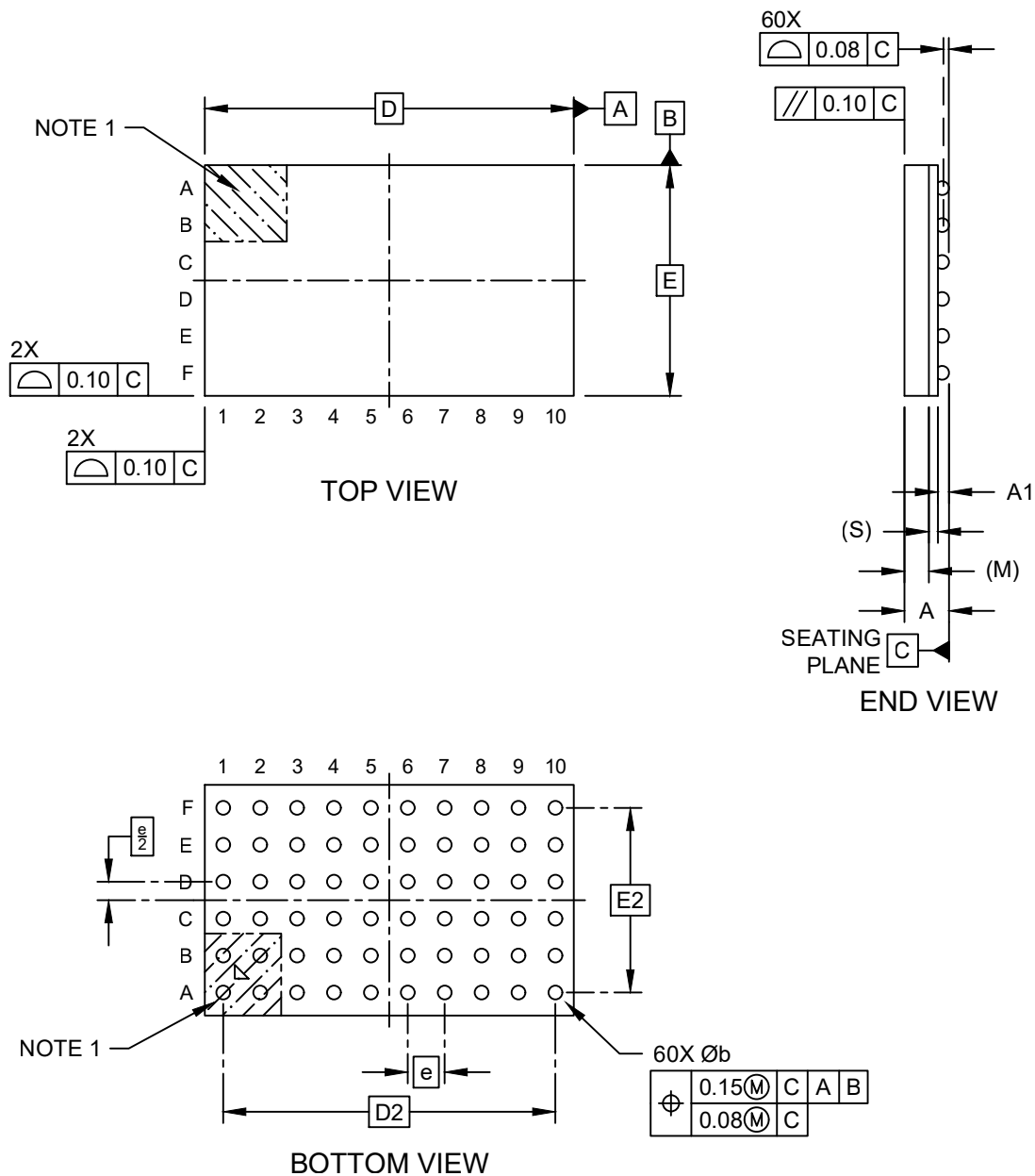


Packaging Diagrams and Parameters

60-Ball Thin Fine-Pitch Ball Grid Array (7GW) - 8x5x1mm Body [TFBGA]

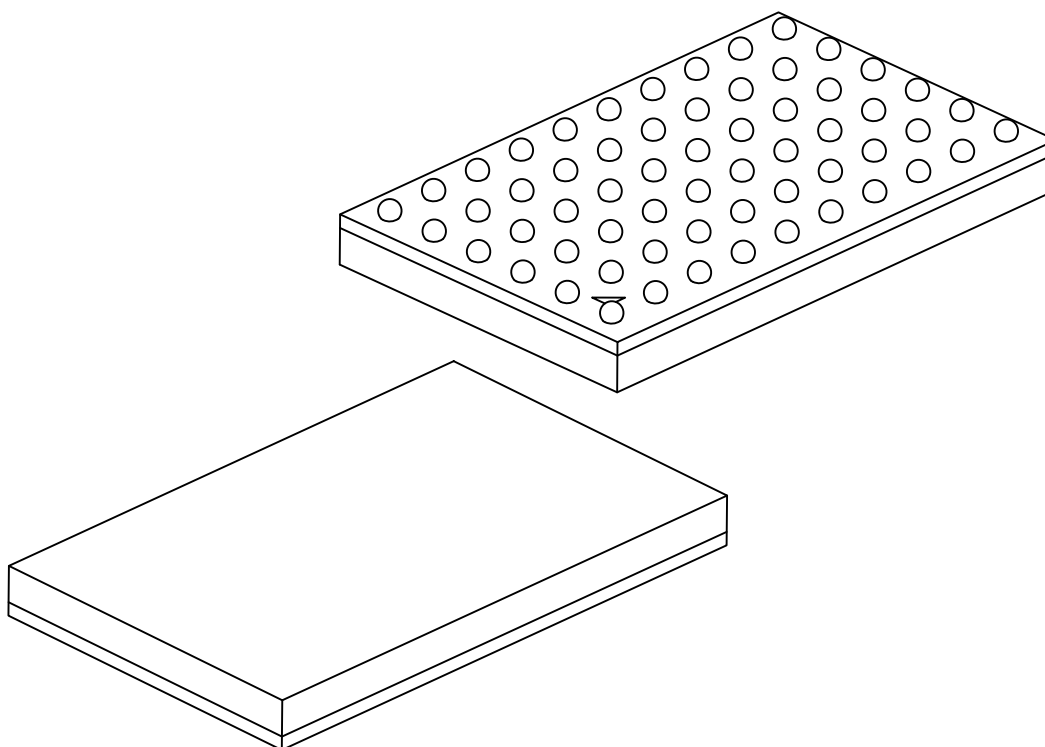
Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



Packaging Diagrams and Parameters

60-Ball Thin Fine-Pitch Ball Grid Array (7GW) - 8x5x1mm Body [TFBGA]

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Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Terminals	N	60		
Pitch	e	0.80 BSC		
Overall Height	A	0.87	0.94	1.01
Ball Height	A1	0.16	0.21	0.26
Mold Thickness	M	0.53 REF		
Substrate Thickness	S	0.20 REF		
Overall Length	D	8.00 BSC		
Ball Array Length	D2	7.20 BSC		
Overall Width	E	5.00 BSC		
Ball Array Width	E2	4.00 BSC		
Ball Diameter	b	0.27	0.32	0.37

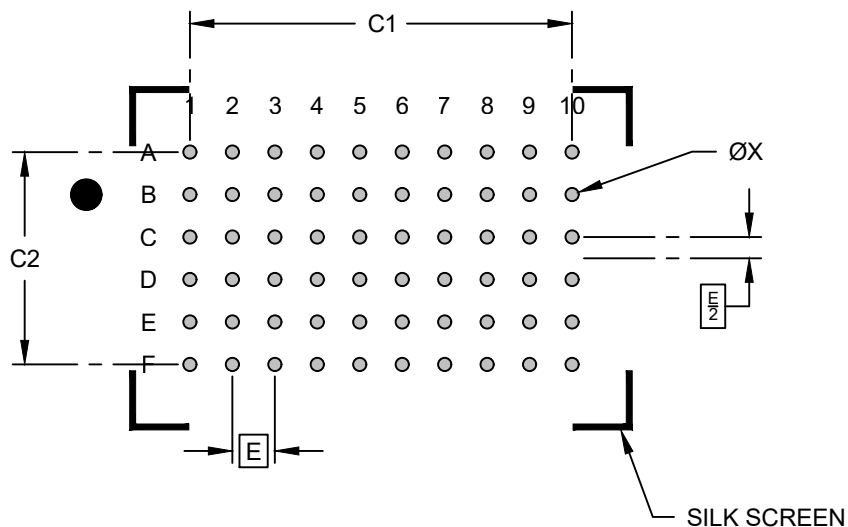
Notes:

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Package is saw singulated
- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
REF: Reference Dimension, usually without tolerance, for information purposes only.

Land Pattern

60-Ball Thin Fine-Pitch Ball Grid Array - (7GW) 8x5x1mm Body [TFBGA]

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E	0.80 BSC		
Contact Pad Spacing	C1		7.20	
Contact Pad Spacing	C2		4.00	
Contact Pad Diameter (X60)	X			0.250

Notes:

- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
- For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during reflow process

Microchip Technology Drawing C04-02646 Rev A